

STORM User Guide

Severe Thunderstorm Observation and Reconnaissance Monitor

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1. Overview

STORM is a standalone desktop application purpose-built for severe weather storm chasing situational awareness. It integrates offline vector maps, real-time NEXRAD radar, GOES satellite imagery, SPC/NWS hazard products, collaborative field annotations, meteorological drawings, and live multi-vehicle tracking — all in a single interface designed to function reliably in the field.

Core Technology Stack:

Component	Technology
UI Framework	PyQt6 (QMainWindow + QWebEngineView)
Map Renderer	MapLibre GL JS (embedded in browser engine)
Vector Tiles	Offline MBTiles served via custom <code>storm://</code> URL scheme
Radar Decoding	MetPy + matplotlib (Level 3, Unidata THREDDS)
Satellite Imagery	IEM WMS (GOES-East, GOES-West)
Hazard Data	SPC GeoJSON MapServer + NWS API
Vehicle/Annotation Sync	MQTT over AWS IoT (TLS)
Python-to-JS Bridge	QWebChannel (<code>bridge</code> object)

Run Command:

```
python main.py
```

2. Installation & Requirements

Prerequisites

- Python 3.10+
- PyQt6 with WebEngine (`pip install PyQt6 PyQt6-WebEngine`)
- MetPy, matplotlib, numpy, scipy (`pip install metpy matplotlib numpy scipy`)
- paho-mqtt (`pip install paho-mqtt`)
- The offline MBTiles vector tile database (`storm.mbtiles`, ~500 MB–1 GB, distributed separately)

MQTT Certificates

Certain network features (vehicle sync, annotation/drawing sync) require AWS IoT TLS certificates. Place them at:

- `/aws/storm.pem` — CA certificate
- `/aws/storm.pem.crt` — device certificate
- `/aws/storm-private.pem.key` — private key

If certificates are missing, STORM starts in a degraded state — map, radar, satellite, and SPC/NWS hazard features remain fully functional, but MQTT-dependent sync features are disabled.

Single-Instance Guard

Only one instance of STORM can run per machine (an internal port is used as a lock). If a second launch is attempted, a warning dialog appears and the second instance exits immediately.

3. Launching STORM

Launch Dialog

Every time STORM starts (unless `--monitor` mode is passed), a configuration dialog appears before the main window opens.

Vehicle ID Your vehicle's identifier on the network (e.g., "Chase1", "Mesonet-TX"). It is pre-filled from the previous session. You can edit it freely. The ID is published with every observation, annotation, and drawing you send to other vehicles.

Data Directory (Track A) An optional folder path for real-time observation files from a local data logger (FOFS truck logger format). If left blank, Track A file input is disabled for the session. Use the "Browse" button to pick a folder. The app will scan for a file matching `YYYYMMDD.txt` for the current date and poll it every 10 seconds.

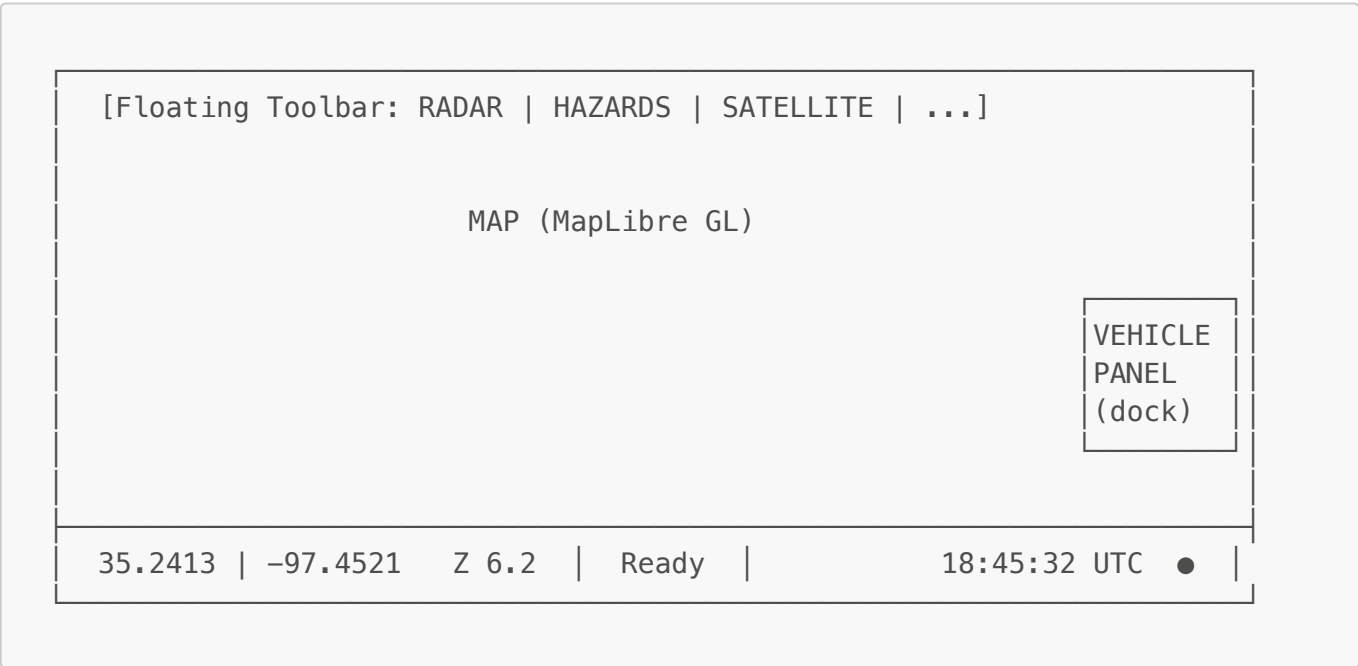
Monitor Mode When checked, local observation inputs (file watcher, GPS) are suppressed. Your vehicle publishes no observations, but you can still view other vehicles' data and control all map overlays. Useful for command post / situational-awareness-only roles.

Update Status During startup, STORM checks for available updates. The dialog will show:

- "Checking for updates..." while polling
- The current and latest version numbers if an update is available
- An "Update Available" button (STORM never auto-updates)

Launch Click Launch to proceed. Your vehicle ID and data directory are saved to persistent settings and pre-filled next time.

4. Main Interface Layout



Region	Description
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Region	Description
Floating Toolbar	Top-left collapsible drawer strip — all overlay and tool controls
Map	Full-window interactive map; all overlays render here
Vehicle Panel	Right-side dock — vehicle list and history graphs
Outlook Panel	Left-side slide-out — SPC/NWS text discussion viewer
Status Bar	Bottom strip — mouse coordinates, zoom, status messages, clock, network indicator

5. Map Navigation

Mouse Controls

Action	Result
Scroll wheel	Zoom in / zoom out
Click + drag	Pan the map
Click a feature	Open info popup (SPC outlook, watch, warning, annotation, etc.)
Right-click	Finish current drawing or measurement
Hover over feature	Show tooltip

Map Layers (Render Order, bottom to top)

1. Offline vector base map (terrain, streets, water bodies, labels)
2. SPC categorical outlook polygons (MRGL / SLGHT / ENH / MDT / HIGH)
3. SPC probabilistic outlook polygons (tornado / wind / hail probability shading + hatching)
4. NWS watches (county-polygon outlines)
5. SPC Mesoscale Discussions (polygon outline + numeric label)
6. NWS warnings (storm-polygon fills, phenom-specific colors)
7. GOES satellite imagery (optional, opacity-controlled)
8. NEXRAD radar overlay (semi-transparent PNG raster)
9. Station plot icons (MetPy-style, at vehicle positions)
10. Vehicle markers (colored dots + info popups)
11. Annotations (road closure / construction / flooding / downed-lines / debris markers)
12. Drawings (fronts, polylines, polygons)
13. Storm motion cones
14. Measurement tool geometry
15. Deployment location markers

6. Floating Toolbar

The floating toolbar lives in the top-left corner of the map window. Each button either toggles a layer directly or expands a collapsible drawer with additional controls. Drawers animate open/closed smoothly.

6.1 Radar

Button: RADAR (checkable toggle)

Clicking the button expands the radar drawer. Checking the checkbox inside enables radar fetching and display.

Drawer Controls

Site Selector (dropdown) Choose from 31 pre-configured NEXRAD sites (e.g., KTLX, KOUN, KVN~~X~~, KFWS, etc.) or type a custom 4-letter site code. Changing the site immediately clears cached frames and begins fetching for the new site.

Product Selector (dropdown)

Option	NEXRAD Product	Description
REFLECTIVITY (SR)	N0B (fallback: N0Q → N0R)	Base reflectivity (dBZ)
VELOCITY	N0U	Base velocity (kt)

Show Data Checkbox Enables or disables radar fetching. When unchecked, radar is hidden from the map but site/product selections are preserved.

Playback Controls

Button	Action
⏮	Jump to oldest cached frame
⏪	Step back one frame
▶ / ⏸	Start / pause auto-playback loop (500 ms per frame)
⏩	Step forward one frame
⏭	Jump to most recent (live) frame
Timeline slider	Scrub to any cached frame by dragging
Time label	Shows current frame's UTC time (HH:MMZ)

How Radar Data Works

- **Source:** Unidata THREDDS Level 3 catalog (public, no authentication required)
- **Polling:** Every 2 minutes
- **Frame cache:** Up to 12 frames per product
- **First fetch:** Backfills 6 reflectivity frames and 6 velocity frames

- **Render pipeline:** Level 3 polar data decoded by MetPy → reprojected to lat/lon grid → rendered as RGBA PNG via matplotlib → base64-encoded → pushed to MapLibre as a raster layer
- **Grid size:** 512×512 by default (configurable via `--render-grid-size`)
- **Render time:** ~80–150 ms on modern hardware at default grid size

Disabling Radar

Pass `--disable-radar` at launch to hide the RADAR button entirely.

6.2 Hazards

Button: `HAZARDS` (checkable toggle)

Clicking the button expands the hazards drawer. The drawer contains two groups of controls.

SPC Products (mutually exclusive — only one active at a time)

Button	Layer
OUTLOOK	Day 1 categorical outlook (MRGL, SLGHT, ENH, MDT, HIGH)
TOR	Tornado probability (2% / 5% / 10% / 15% / 30% / 45% / 60%)
WIND	Wind probability (5% / 15% / 30% / 45% / 60%)
HAIL	Hail probability (5% / 15% / 30% / 45% / 60%)

Selecting one clears the others. Clicking the already-active button deactivates it (all SPC probability layers hidden).

Additive Overlays (independent, stack on top of each other)

Button	Layer
WATCHES	SPC Tornado and Severe Thunderstorm Watch county polygons
MDs	SPC Mesoscale Discussion polygons with number labels
NWS WARNINGS	Active NWS warnings (all VTEC phenomenons)

These can be enabled in any combination, independent of the SPC product selection.

Legend

When any hazard product is active, a compact color legend row appears below the drawer controls showing color swatches and probability labels. It animates in and out with the drawer.

Clicking Hazard Features

SPC Categorical / Tornado / Wind / Hail polygons:

- Click a polygon → popup with risk level and probability label

- Click "Read Discussion" in the popup → fetches full text from IEM/SPC and displays it in the [Outlook Panel](#)

Watches:

- Click a county-polygon → popup with watch number, type (Tornado or Severe Thunderstorm), and issuance details
- Click "Read" → fetches full SEL bulletin text and displays it in the Outlook Panel

Mesoscale Discussions:

- Click an MD polygon → popup with MD number
- Click "Read" → fetches full MD text from SPC and displays it in the Outlook Panel

NWS Warnings:

- Click a warning polygon → popup with product type, event number, and NWS headline
- Full description and instructions fetched from the NWS API and shown in the popup

Significant Area Hatching

SPC tornado, wind, and hail probability layers include significant (CIG/SIGN) areas as separate overlapping features rendered with gray fill plus hatch patterns:

Pattern	Meaning
Dashed diagonal lines	Conditional Intensity Group 1 (CIG1)
Solid diagonal lines	CIG2 or SIGN (significant tornado area)
Checkerboard	CIG3

Color Reference

SPC Categorical:

Category	Color
MRGL (Marginal)	Green
SLGHT (Slight)	Yellow
ENH (Enhanced)	Orange
MDT (Moderate)	Red
HIGH	Purple

Watches:

Watch Type	Color
Tornado Watch	Red (#FF0000)

Watch Type	Color
Severe Thunderstorm Watch	Royal Blue (#4169E1)

NWS Warnings by VTEC Phenom:

Phenom	Event	Color
TO	Tornado Warning	Red (#FF0000)
SV	Severe Thunderstorm Warning	Gold (#FFD700)
FF	Flash Flood Warning	Green (#00FF00)
FL	Flood Warning	Cyan-green (#00FF7F)
WS	Winter Storm Warning	Pink (#FF69B4)
BZ	Blizzard Warning	Orange-red (#FF4500)

Mesoscale Discussions: Magenta (#FF66CC) outline

Data Polling

Product	Interval	Cache TTL
SPC Categorical + Probabilities	15 min	15 min
SPC Watches	2 min	None (always fresh)
SPC MDs	2 min	None
NWS Warnings	2 min	None

When you toggle a product on within the cache window, data displays instantly from cache. After TTL expiry, a fresh fetch is triggered automatically.

6.3 Satellite

Button: **SATELLITE** (checkable toggle)

Clicking the button expands the satellite drawer.

Coverage Mode Selection

Button	Description
CONUS	Full continental US view (1600×800 px, updates every 5 min)
MESO-1	Mesoscale sector 1 (1024×1024 px, updates every 1 min)
MESO-2	Mesoscale sector 2 (1024×1024 px, updates every 1 min)

MESO-1 and MESO-2 buttons are disabled until the satellite system confirms those sectors are available. Hover over MESO-1 or MESO-2 buttons to preview the sector boundary on the map as a semi-transparent overlay.

Opacity Slider

Controls satellite image transparency from 0% (invisible) to 100% (fully opaque). Default is 70%. Useful for seeing underlying map features through the imagery.

Playback Controls

Identical layout to the radar playback row:

Button	Action
⏮	Jump to oldest frame
⏪	Step back one frame
▶ / ⏸	Start / pause auto-playback (500 ms per frame)
⏩	Step forward one frame
⏭	Jump to newest frame
Timeline slider	Scrub to any cached frame
Time label	Current frame UTC time (---:--Z if no data yet)

How Satellite Data Works

- **Source:** IEM GOES-East WMS (mesonet.agron.iastate.edu) serving current operational GOES satellite (GOES-19 as of 2025)
- **Request format:** WMS GetMap PNG
- **Frame cache:** Up to 10 frames per mode
- **CONUS polling:** Every 5 minutes; 10 frames backfilled on mode selection
- **MESO polling:** Every 1 minute once sectors are confirmed available
- **Memory usage:** CONUS ~10–20 MB; each MESO sector ~20–30 MB

6.4 Annotations

Button: [ANNOTATIONS](#) (checkable toggle, expands drawer)

Annotations are field-condition markers placed on the map and synced to all vehicles over MQTT.

Annotation Types

Icon	Type	Color	Use For
×	Road Closure	Red (#E53935)	Blocked road — no through traffic
▲	Construction	Yellow (#FFD166)	Active road work

Icon	Type	Color	Use For
~	Flooded Road	Blue (#4A9EFF)	Standing water over road
	Downed Power Lines	Yellow (#FFD166)	Electrical hazard on road
!	Road Debris	Orange (#FF6B35)	Debris field on road

Placing an Annotation

- 1. Click the desired annotation type button in the drawer — it highlights in that type's color.
- 2. Click any location on the map.
- 3. A dialog opens showing the clicked coordinates and an optional "Note" text field.
- 4. Add any relevant note (e.g., "Road completely blocked at intersection with CR-1234").
- 5. Click **Add to Map**.
- 6. The annotation marker appears on the map immediately and is published to all connected vehicles via MQTT.

Editing or Deleting an Annotation

- 1. Click an existing annotation marker on the map.
- 2. The same dialog opens, pre-filled with the existing note.
- 3. Edit the note or click **Delete** to remove it.
- 4. Click **Save** to confirm.

Changes and deletions are published to all connected vehicles.

6.5 Drawings

Button: DRAWINGS (collapsible drawer)

Drawings are meteorological features sketched on the map — synced to all vehicles over MQTT.

Front Types

Button	Front	Symbol	Color
Cold Front	Cold Front	Triangles pointing in direction of motion	Blue (#4A9EFF)
Warm Front	Warm Front	Semicircles pointing in direction of motion	Red (#E53935)
Stationary Front	Stationary Front	Alternating triangles + semicircles on one line	Blue + Red
Occluded Front	Occluded Front	Alternating triangles + semicircles, same side	Purple (#9C27B0)
Dryline	Dryline	Scalloped arcs above the line	Brown (#D4872E)

Custom Shapes

Button	Shape	Color
—	Polyline	White (#E8EAF0)
□	Polygon	White (#E8EAF0)

Drawing a Front or Shape

1. Click the front type or shape button — it highlights and the map enters drawing mode (crosshair cursor).
2. Click the map to place the first point, then continue clicking to trace the line.
3. **Finish:** Right-click anywhere or press **Escape**.
4. A dialog prompts for a title (optional for fronts; required for polylines and polygons; defaults to the type name if left blank).
5. The feature appears on the map with the appropriate symbol and a title label at its center.

Editing or Deleting a Drawing

1. Click an existing drawing on the map.
2. The dialog opens with the current title.
3. For fronts: an option to flip direction (reverses the symbol orientation) is available.
4. Edit the title, flip direction, or click **Delete**.
5. Click **Save** to confirm.

All changes are synced via MQTT.

Disabled If

`--disable-annotations` is passed at launch — this flag disables annotations, drawings, station plots, and the storm motion cone together.

6.6 Storm Motion Cone

Button: `CONE`

Places a storm motion cone on the map — a forward-propagating sector showing the expected storm track with a 20° angular spread and time rings.

Creating a Cone

1. Click the **CONE** button.
2. Click any location on the map (typically the current storm location).
3. A dialog opens with two fields:
 - **Speed (knots):** Expected storm forward speed. Default 35 kts. Range: 0–200.
 - **Heading (degrees):** Direction the storm is moving *toward*. Default 240° (toward the SSW). Range: 0–359.
4. Click **Add to Map**.

The rendered cone includes:

- A base point at the clicked location
- Two radial bounding lines extending 1 hour of travel at $\pm 20^\circ$ spread
- Concentric time rings at 0.25, 0.5, 0.75, and 1.0 hour
- A center spine line showing the primary motion direction

Color: Cyan (#00CFFF)

Editing or Deleting a Cone

1. Click the cone on the map.
2. The dialog shows the current speed and heading.
3. Update the values or click **Delete**.
4. Click **Save**.

Cones are synced via MQTT (`storm/cones/{id}` topic).

Disabled If

`--disable-annotations` is passed at launch.

6.7 Measurement Tool

Button: `MEASURE` (checkable toggle)

Measures great-circle distances on the map.

Usage

1. Click the **MEASURE** button — the map cursor changes to a crosshair.
2. Click a start point on the map — a dot appears.
3. Move the mouse — a preview line and live distance label follow the cursor.
4. Click an end point — the segment is fixed.
5. Continue clicking to measure additional segments (multi-segment path).
6. **Finish:** Right-click or press **Escape**.
7. Distance is shown in miles (e.g., "47.3 mi") with total path length.

Clear: A **Clear** button removes all measurement geometry from the map.

Color: Cyan lines and labels

6.8 Station Plots

Button: `STATION PLOTS` (checkable toggle)

Displays MetPy-style meteorological station plot icons at each tracked vehicle's current position.

Station Plot Layout

[Temp °F] [Pressure code]

○—wind barb

[Dewpt °F]

Position	Data	Color
Upper-left (NW)	Temperature (°F)	Red text
Lower-left (SW)	Dewpoint (°F)	Green text
Upper-right (NE)	Pressure code (last 3 digits × 10, e.g., 1013.2 mb → "132")	White text
Center	Wind barb (meteorological convention, barbs point toward station)	White
Center dot	Anchor	White circle

Wind Calm: If wind speed is ≤2 kt, only a small circle is rendered (no barbs).

How Station Plots Are Rendered

Station plots are generated as PNG images (135×135 px, transparent background) by MetPy and matplotlib, then pushed to the map as image markers. Each vehicle's plot is cached and only re-rendered when new observation data arrives.

Disabled If

`--disable-annotations` is passed at launch.

6.9 Deployment Locations

Button: `DEPLOY LOCS` (checkable toggle)

Displays historical deployment locations from a static JSON file as markers on the map.

- **Data source:** `data/deployment_locations.json`
- **Format:** JSON array of `{lat, lon, label, date}` objects
- **Display:** Small dot/icon markers; hover to see label and date in a tooltip
- Toggle visibility without reloading — markers are pre-loaded at startup

Disabled If

`--disable-deploy-locs` is passed at launch, or the locations file is missing.

6.10 Vehicle Panel

Button: `VEHICLES` (toggles the right-side dock panel)

The vehicle panel provides a live roster of all tracked vehicles and time-series observation graphs.

Vehicle List (upper section)

Each vehicle row shows:

Field	Description
ID	Vehicle identifier (e.g., "Chase1")
Position	Compact lat/lon (e.g., "35.24
Last Update	Relative time (e.g., "just now", "2 min ago")
Color dot	Matches the vehicle's marker color on the map

Click a vehicle row to **zoom the map to that vehicle's location**.

Vehicles are sorted by most-recently-updated first.

History Graph (lower section)

Selecting a vehicle (or when your own vehicle is the only one) shows a 4-panel time-series chart of the last 10 minutes of observations:

Panel	Data	Color
Top	Temperature (°F)	Orange line with dots
2nd	Dewpoint (°F)	Blue line with dots
3rd	Wind Speed (knots)	Green line with dots
Bottom	Pressure (mb/hPa)	Yellow line with dots

The bottom panel shows a UTC time axis (HH:MM format). Subtle gray grid lines span all panels. The chart updates in real-time as new observations arrive.

7. Status Bar

The status bar runs along the bottom of the main window.

Section	Content
Left	Mouse coordinates (lat/lon) and zoom level, updated live as you move the cursor over the map. Example: 35.2413 -97.4521 Z 6.2
Center	Status messages: "Ready" (default), "MQTT connected", error messages (e.g., "Radar fetch error: timeout")
Right	Current UTC clock (updates every second). Example: 18:45:32 UTC
Far right	Network status indicator pill: Green = Online, Yellow = Slow (>1 s response), Red = Offline. Checked every 30 seconds via Cloudflare DNS.

8. Outlook Text Panel

The Outlook Panel is a left-side slide-in drawer that displays the full text of SPC and NWS products.

Opening the Panel: Click any SPC feature (outlook area, watch polygon, or MD polygon) on the map, then click "Read" or "Read Discussion" in the popup.

What Appears:

- **Title bar:** Product name (e.g., "DAY 1 CONVECTIVE OUTLOOK", "MD 0179", "TORNADO WATCH 0029")
- **Close button (x):** Slides the panel back out
- **Text area:** Full monospaced text of the product (read-only, scrollable)

Text Sources:

Product	Source
Day 1 Outlook	IEM AFOS (<code>pil=SWODY1</code>)
Mesoscale Discussions	SPC direct (<code>spc.noaa.gov/products/md/md{NNNN}.txt</code>)
Tornado Watches	IEM AFOS (<code>pil=SEL{n}</code> where <code>n = watch_number % 10</code>)
SVR Tstm Watches	IEM AFOS (same PIL logic as tornado)
NWS Warnings	NWS API JSON (<code>headline</code> + <code>description</code> + <code>instruction</code> fields)

Animation: Slides in from the left (200 ms), slides out on close (200 ms). Panel is 340 px wide.

9. Vehicle Tracking & Observations

STORM tracks vehicles through several parallel data input channels. All inputs feed into a common observation store that drives the vehicle panel, station plots, and map markers.

Track A — Local File Watcher

Polls a FOFS truck logger format file every 10 seconds. Enabled only if a data directory was selected at launch.

Expected filename: `YYYYMMDD.txt` (today's date)

Expected columns (tab or comma delimited):

Column	Format	Description
<code>gps_date</code>	DDMMYY	GPS date
<code>gps_time</code>	HHMMSS	GPS time (UTC)
<code>lat</code>	decimal degrees	Latitude
<code>lon</code>	decimal degrees	Longitude

Column	Format	Description
<code>t_fast</code>	°F	Fast-response temperature
<code>dewpoint</code>	°F	Dewpoint temperature
<code>sfc_wspd</code>	knots	Surface wind speed
<code>sfc_wdir</code>	degrees	Surface wind direction
<code>pressure</code>	mb	Atmospheric pressure

Track B — GPS Reader

Automatically detects and reads NMEA sentences from a connected serial GPS device. Provides position-only data (no met obs). Active only if `--disable-data-inputs` is not passed.

MQTT — Inbound Vehicle Observations

Receives observations from other vehicles via the `storm/vehicles/{id}` topic. Active whenever MQTT is connected and not disabled.

NSSL External Vehicle Fetcher

Polls an external NSSL JSON feed every 10 seconds to display the locations of other chase vehicles on the map. These vehicles show as markers but do not have history graphs (no obs data, position only).

URL: `https://data.nssl.noaa.gov/thredds/fileServer/F0FS/Mobile-Mesonet/placefile_info/storm_locs.json`

Disabled by: `--disable-vehicle-fetcher`

Observation History Buffer

Each vehicle maintains a 10-minute rolling buffer of observations. This buffer drives the history graph in the vehicle panel. The buffer is kept in memory only — it does not persist across STORM restarts.

10. Network & MQTT Sync

STORM uses MQTT over AWS IoT (TLS) to synchronize annotations, drawings, cones, and vehicle observations across all connected vehicles in the network.

Topics

Topic	Content
<code>storm/vehicles/{id}</code>	Vehicle observations (lat, lon, temp, dewpoint, wind, pressure)
<code>storm/annotations/{id}</code>	Road condition annotations (create / edit / delete)
<code>storm/drawings/{id}</code>	Meteorological drawings (fronts, polylines, polygons)
<code>storm/cones/{id}</code>	Storm motion cones

Behavior

- On connect, retained messages for all active annotations, drawings, and cones are received and displayed automatically.
- Disconnect/reconnect is handled automatically — STORM continuously attempts to reconnect with exponential backoff.
- If `--disable-mqtt` is passed, all sync features are disabled and the vehicle panel shows only locally-observed vehicles.
- Status bar shows "MQTT connected" or an error message.

11. Data Sources & Polling Intervals

Data Source	Polling Interval	Notes
NEXRAD Radar (Unidata THREDDS)	Every 2 minutes	Level 3 products; no auth required
SPC Categorical Outlook	Every 15 minutes	Day 1 only
SPC Tornado / Wind / Hail Probabilities	Every 15 minutes	All fetched together with categorical
SPC Watches	Every 2 minutes	County polygons, tornado + SVR tstm
SPC Mesoscale Discussions	Every 2 minutes	MD polygon geometry + number
NWS Warnings	Every 2 minutes	All active VTEC phenomenons
GOES Satellite (CONUS)	Every 5 minutes	Full CONUS, up to 10 frames
GOES Satellite (MESO-1/2)	Every 1 minute	When active, per-sector
NSSL Vehicle Feed	Every 10 seconds	External chase vehicle positions
Local Obs File (Track A)	Every 10 seconds	Today's YYYYMMDD.txt
GPS (Track B)	Real-time / continuous	NMEA sentences
MQTT Inbound	Real-time	Vehicle obs, annotations, drawings, cones
Network Health Check	Every 30 seconds	Cloudflare 1.1.1.1 ping

12. Command-Line Options

Run `python main.py --help` for the full list. Key options:

General

<code>--debug</code> (Ctrl+D)	Enable debug logging and the debug panel
<code>--log-level LEVEL</code>	Logging verbosity: DEBUG, INFO, WARNING,

<code>ERROR (default: WARNING)</code> <code>--monitor</code>	Monitor mode: suppresses local obs inputs
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Disable Features

<code>--disable-radar</code> <code>fetching</code>	Hide the RADAR control; disable all radar
<code>--disable-mqtt</code> <code>obs)</code>	Disable all MQTT features (sync, vehicle
<code>--disable-vehicle-fetcher</code> <code>polling</code>	Disable NSSL external vehicle location
<code>--disable-annotations</code> <code>plots, and cones</code>	Disable annotations, drawings, station
<code>--disable-deploy-locs</code>	Disable deployment location markers
<code>--disable-data-inputs</code>	Disable local file watcher and GPS reader

Replay Mode

<code>--truck-replay-file PATH</code> <code>live data</code>	Replay a local obs CSV/TXT file instead of
<code>--truck-replay-interval-ms MS</code> <code>(default: 1000 ms)</code>	Interval between replayed samples
<code>--truck-replay-restamp</code> <code>"now"</code>	Shift timestamps so the last obs lands at

Performance

<code>--render-grid-size N</code> <code>(default), 768, or 1024</code>	Radar rendering grid: 128, 256, 512
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Debug Run Profiles

<code>--debug-run N</code> <code>size)</code>	Quick diagnostic profile (0-6): 0 = full features 1 = disable MQTT only 2 = safe mode (minimal features, low grid
<code>--enable-startup-toggles</code> <code>debug-run profile</code>	3-6 = various feature subsets for testing Allow CLI disable flags to override the

Network

--mqtt-no-tls

MQTT without TLS (development use only, not for field use)

13. Configuration & Certificates

config.py

Edit `config.py` for persistent defaults:

Constant	Description
ACCENT_COLOR	UI accent color (default: "#00CFFF", cyan)
VEHICLE_ID	Default vehicle ID (overridden by launch dialog)
OBS_FILE_DIR	Default data directory path (overridden by launch dialog)
MQTT_HOST	AWS IoT endpoint hostname
MQTT_PORT	MQTT port (default: 8883 for TLS)
MQTT_USE_TLS	Whether to require TLS (default: True)
VEHICLES_URL	NSSL external vehicle location feed URL

Persistent Session State (QSettings)

The following are saved automatically and restored at next launch:

- Window geometry and dock layout
- Last-used vehicle ID
- Last-used data directory

Settings are stored under the `"NSSL" / "STORM"` namespace in your OS's native settings storage.

TLS Certificate Files

Place these three files exactly at these paths before using MQTT:

File	Role
/aws/storm.pem	CA certificate
/aws/storm.pem.crt	Device certificate
/aws/storm-private.pem.key	Private key

Missing certificates cause MQTT to fail silently — all other features remain operational.

14. Keyboard Shortcuts

Key	Action
Escape	Cancel active drawing / annotation placement / measurement mode
Ctrl+D	Toggle debug panel
Ctrl+E	Toggle error log panel
Right-click	Finish drawing or measurement in progress

15. Performance Tuning

Radar Grid Size

The radar rendering grid controls the sharpness and rendering time of the NEXRAD overlay.

Grid Size	Render Time (approx.)	Recommended Use
128	~5–10 ms	Very slow hardware, emergency fallback
256	~20–40 ms	Older laptops, safe-mode deployments
512	~80–150 ms	Default — balanced quality/speed
768	~200–400 ms	High-end hardware, sharper image
1024	~400–800 ms	Workstation-class machines only

Pass `--render-grid-size N` to override. Lower grid sizes make the radar appear blockier but keep the CPU free for other tasks.

Memory Usage

Component	Approximate Memory
Radar frame cache (12 frames)	20–50 MB
CONUS satellite cache (10 frames)	10–20 MB
MESO satellite cache (10 frames/sector)	20–30 MB per sector
Station plot image cache	~1–5 MB (per vehicle)
Observation history buffer (10 min/vehicle)	Negligible

Safe Mode

If the application is unstable on older hardware (WebGL context loss, slow renders), launch with:

```
python main.py --debug-run 2
```

Safe mode reduces grid size to 128, disables some overlays, and minimizes background polling.

16. Diagnostics & Error Handling

Debug Mode

```
python main.py --debug
```

Enables:

- Verbose logging to console (DEBUG level)
- The debug panel (also toggled with **Ctrl+D** at any time)
- Fault handler writing stack traces to `storm_fault.log` on crash

Error Log Panel

Press **Ctrl+E** at any time to open the error log panel. It shows all WARNING-level and above log messages captured since launch. This log is also written to `storm_errors.log` and survives across restarts.

Network Status Indicator

The colored pill in the bottom-right of the status bar updates every 30 seconds:

- **Green (Online):** Connectivity to Cloudflare 1.1.1.1 is fast
- **Yellow (Slow):** Response time exceeded 1 second — data fetches may be delayed
- **Red (Offline):** No connectivity — no real-time data will update

Single-Instance Guard

STORM uses an internal TCP port (19876) as a process lock. If a second instance is launched while one is already running, a dialog warns the user and the second instance exits.

Common Issues

Symptom	Likely Cause	Resolution
Radar not updating	THREDDDS server unavailable or slow	Check network indicator; try a different site
"MQTT disconnected" in status bar	Certificate missing or endpoint unreachable	Verify <code>/aws/</code> certificate files exist
Station plots missing	<code>--disable-annotations</code> passed	Relaunch without that flag
MESO-1/2 buttons grayed out	Sectors not currently available	Wait — buttons enable automatically when sectors are active
Map tiles blank/gray	<code>storm.mbtiles</code> missing or corrupted	Verify the MBTiles file is present and not zero-byte

Symptom	Likely Cause	Resolution
Slow/choppy radar rendering	Grid size too large for hardware	Use <code>--render-grid-size 256</code>
Hatch patterns not visible	WebGL limitation	Known quirk; patterns are built as pixel data, not canvas — should render in all supported modes

17. Known Limitations & Quirks

Canvas Readback MapLibre GL running inside QWebEngineView cannot reliably read back canvas pixel data. Hatch patterns (CIG/SIGN significant areas) are therefore built entirely as raw `Uint8Array` pixel data and added to MapLibre via `addImage()`, rather than drawn on a canvas. This is a known architectural constraint.

QWebChannel Async Initialization The `bridge` object that connects Python and JavaScript is initialized asynchronously after the map loads. All JavaScript API functions are stubbed as no-ops (`_stormNoop`) until the bridge is ready. In practice this is transparent to the user, but very rapid actions in the first second after launch may be silently dropped.

MBTiles File Not Included in Repository The offline vector tile database (`storm.mbtiles`) is large (500 MB–1 GB) and is not included in the git repository. It must be downloaded separately and placed in the project root.

SPC Text Product Latency IEM AFOS sometimes responds slowly (up to 20 seconds per request). STORM waits patiently rather than timing out aggressively. This is normal behavior during high-traffic SPC issuance periods.

MESO Sectors Availability GOES MESO-1 and MESO-2 sectors are not always active or available over your area of interest. STORM automatically enables/disables the MESO buttons based on sector availability. This is a function of NOAA's operational GOES scanning schedule.

GOES Satellite Band The IEM WMS endpoint serves the default operational visible or IR band for the current GOES satellite. Band selection is not user-configurable from within STORM.

SPC Data Lag SPC GeoJSON products (tor, wind, hail) are typically updated once or twice daily. They may not reflect very recent issuances until the next polling cycle completes (up to 15 minutes after the fact). SPC Watches and MDs poll every 2 minutes and are effectively near-real-time.

18. Feature Availability Matrix

Feature	Default	Disabled By
Offline vector base map	✔ Always	n/a (requires storm.mbtiles)
NEXRAD Radar overlay	✔ Enabled	<code>--disable-radar</code>
GOES Satellite overlay	✔ Enabled	n/a

Feature	Default	Disabled By
SPC Hazards (outlook, tor, wind, hail)	✓ Enabled	n/a
SPC Watches	✓ Enabled	n/a
SPC Mesoscale Discussions	✓ Enabled	n/a
NWS Warnings	✓ Enabled	n/a
Outlook Text Panel	✓ Enabled	n/a
Annotations (road conditions)	✓ Enabled	--disable-annotations
Drawings (fronts, polylines, polygons)	✓ Enabled	--disable-annotations
Storm Motion Cone	✓ Enabled	--disable-annotations
Station Plots	✓ Enabled	--disable-annotations
Measurement Tool	✓ Enabled	n/a
Deployment Locations	✓ Enabled	--disable-deploy-locs or missing data file
Vehicle Panel	✓ Enabled	n/a
MQTT Sync (annotations, drawings, cones)	✓ Enabled	--disable-mqtt or --disable-annotations
MQTT Vehicle Obs (inbound)	✓ Enabled	--disable-mqtt
Local File Watcher (Track A)	✓ Enabled (if dir set)	--disable-data-inputs or no dir selected
GPS Reader (Track B)	✓ Enabled (if device present)	--disable-data-inputs
NSSL External Vehicle Fetcher	✓ Enabled	--disable-vehicle-fetcher
Radar Playback (loop mode)	✓ Enabled	--disable-radar
Satellite Playback (loop mode)	✓ Enabled	n/a

STORM — Severe Thunderstorm Observation and Reconnaissance Monitor Internal field operations tool — NSSL Mobile Mesonet program